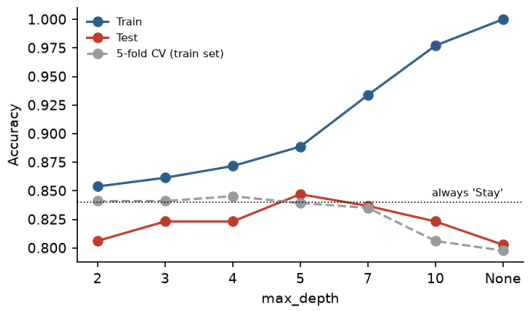
# Predicting employee attrition with decision trees

Ignacio Queipo de Llano · Predictive Machine Learning in Finance, IESE MiF · 12 November 2026

**Bottom line.** Only 16.1% of the 1,470 employees left, so a model that always says "Stay" is already 84% accurate. The best tree (depth 5) reaches 84.7% on the test set and 84.1% in cross-validation: *in accuracy it barely beats doing nothing.* Its value is elsewhere: it finds a third of the leavers with few false alarms and turns them into rules a manager can read. Logistic regression is better on every metric (87.8% accuracy, 40% of leavers caught, AUC 0.83 vs 0.67). Recommendation: score with the logit, lower the alarm to 25% (catches 68% of leavers), and use a depth-3 tree to explain *who* is at risk.

**Q1–Q2. Data and variable types.** 35 columns: the target, 7 text (nominal: `Department`, `JobRole`, `MaritalStatus`…; binary: `OverTime`, `Gender`), 9 integer codes that are really *ordinal* categories (satisfaction 1–4, `Education`, `JobLevel`, `StockOptionLevel`), 14 true numbers (income, age, years), and 4 useless ones dropped (3 constant, 1 ID). Strongest raw links: `JobRole` (Sales Representatives 40% leave vs 2.5% Research Directors), `OverTime` (31% vs 10%), `JobLevel` 1 (26%), `Single` (26%), low `JobInvolvement` (34%); leavers are younger, earlier in their career and paid less (about 0.45 s.d. below stayers). *Does the type matter for a tree?* Barely: a tree only asks "is $x \leq$ threshold?", so an ordinal code works as a number. Text must be encoded: `OverTime` as 0/1, `BusinessTravel` as 0/1/2 (it has an order), `MaritalStatus` and `Department` one-hot (no order). 20 features, 24 columns; stratified 80/20 split (1,176 / 294, 16% leavers in both).




**Q3. Depth.** Train accuracy rises to 100% with no limit (151 leaves: it memorises). Test accuracy peaks at **depth 5 (84.7%)** and falls from there: overfitting starts after 5. But the curve is flat between 2 and 5, and redoing the split with 50 seeds, depth 2 or 3 wins 34 times and depth 5 only 8: "5" is partly luck of this split.

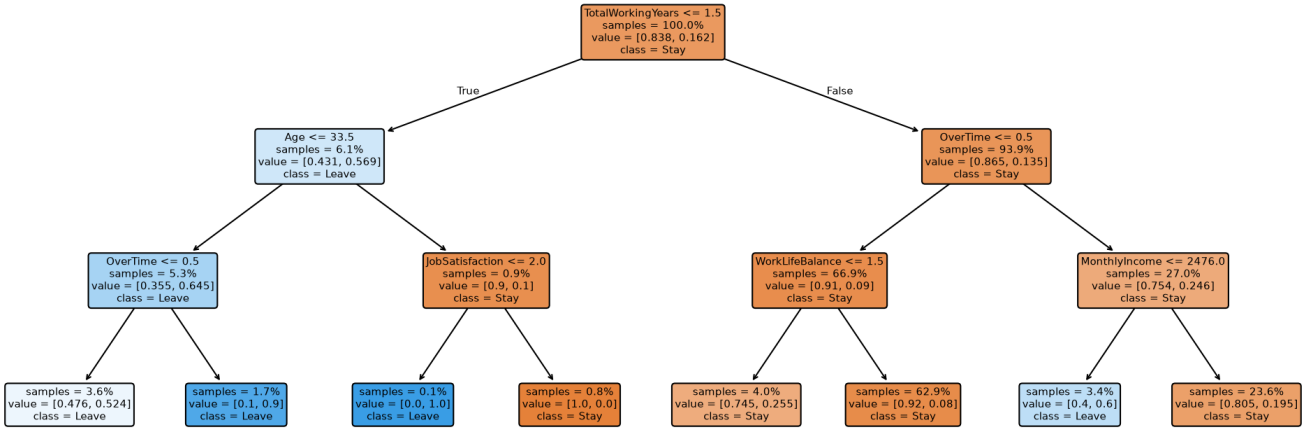
**Q4. Cross-validation (depth 5).** Folds 82.3–85.7%, mean **84.1%**, s.d. **1.3 pp**. Slightly below the single test score (84.7%), which was the lucky draw we picked the depth on. The s.d. is small, so the model is stable, but 1.3 pp is six times its edge over "always Stay" (83.9%, a 0.2 pp gap).

**Q6. Importances.** `TotalWorkingYears` 0.27, `OverTime` 0.13, `MonthlyIncome` 0.12, `Age` 0.09, `Single` 0.06. Matches intuition (early career, overworked, underpaid). Satisfaction scores matter little in the tree but a lot in the logit: the tree uses a few strong splits, the logit adds up many small effects.

**Q5 and Q8. Confusion matrix and model comparison** (test set, 294 employees, 47 leavers):

| | TP | FN | FP | TN | Accuracy | TP rate (recall) | FP rate | AUC |
|---|---|---|---|---|---|---|---|---|
| Always "Stay" | 0 | 47 | 0 | 247 | 84.0% | 0% | 0% | 0.50 |
| Tree, depth 5 | 16 | 31 | 14 | 233 | 84.7% | 34.0% | 5.7% | 0.67 |
| Logistic regression | 19 | 28 | 8 | 239 | **87.8%** | 40.4% | **3.2%** | **0.83** |
| Logit, alarm at 25% | 32 | 15 | 36 | 211 | 82.7% | **68.1%** | 14.6% | 0.83 |

*HR reading.* A **false negative** is a leaver we missed: replacing them costs 50–200% of a salary. A **false positive** is a stayer we flagged: the cost is a retention conversation or bonus spent on someone who was not leaving. Missing a leaver is far more expensive, so accuracy at the default 50% cut-off is the wrong target: the tree misses 2 of every 3 leavers. Lowering the alarm trades cheap false positives for expensive false negatives.




**Q7. Business rules (depth-3 tree).** Each box: share of employees, [stay, leave] proportions.

1. **If** total working years $\leq 1$ **and** age $\leq 33$ **and** overtime: **90% leave** (20 people).
2. **If** more than 1 year of experience **and** overtime **and** monthly income $\leq$ \$2,476: **60% leave** (40 people).
3. **If** more than 1 year of experience, no overtime and work-life balance not "Bad": **8% leave** — the safe 63% of the workforce. (One leaf holds a single employee: that is noise, not a rule.)

**Q8. Recommendation.** Use the **logistic regression** to score everyone: higher accuracy, more leavers caught, fewer false alarms, and it ranks risk much better (AUC 0.83). With the alarm at 25% HR reaches 68% of leavers. Use the **depth-3 tree** to explain the drivers in a meeting (overtime, first job, low pay); a tree deep enough to compete is 24 leaves and no longer readable.